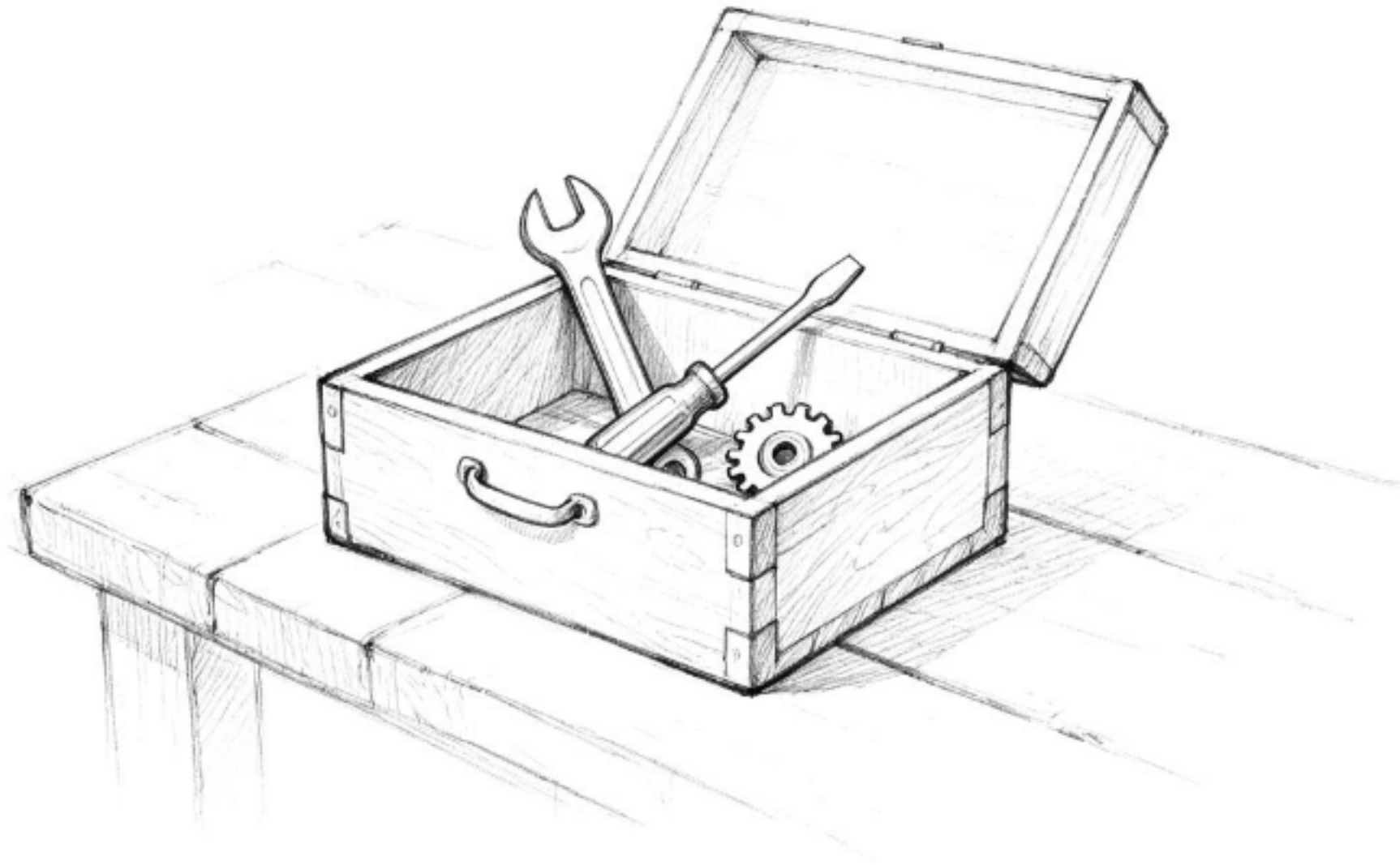
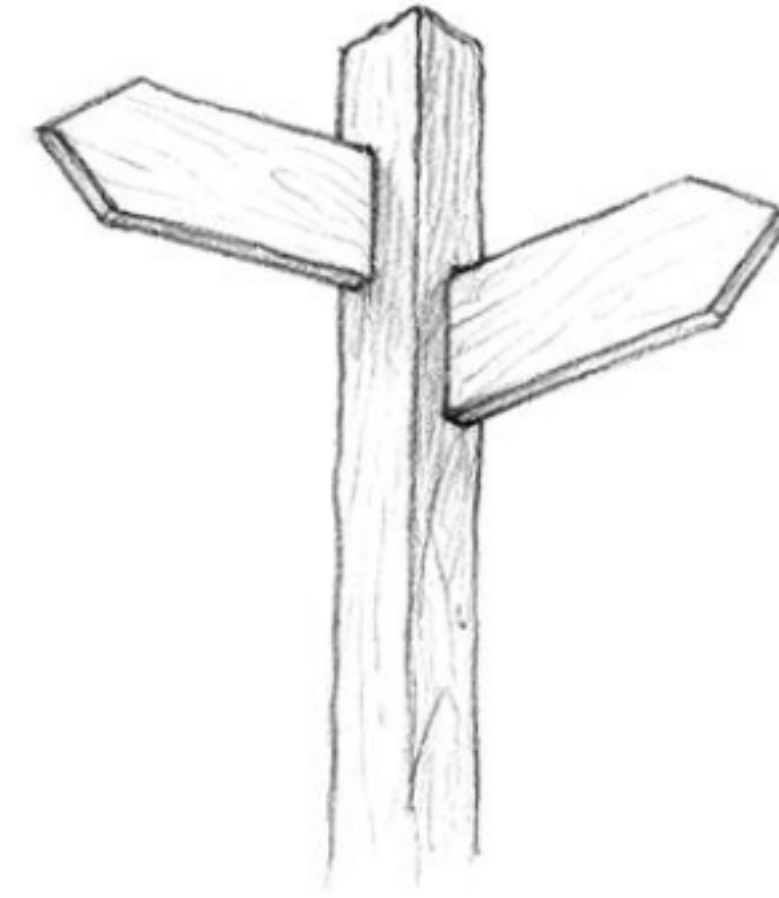


The ML Algorithm Toolbox — The AI You Don't See



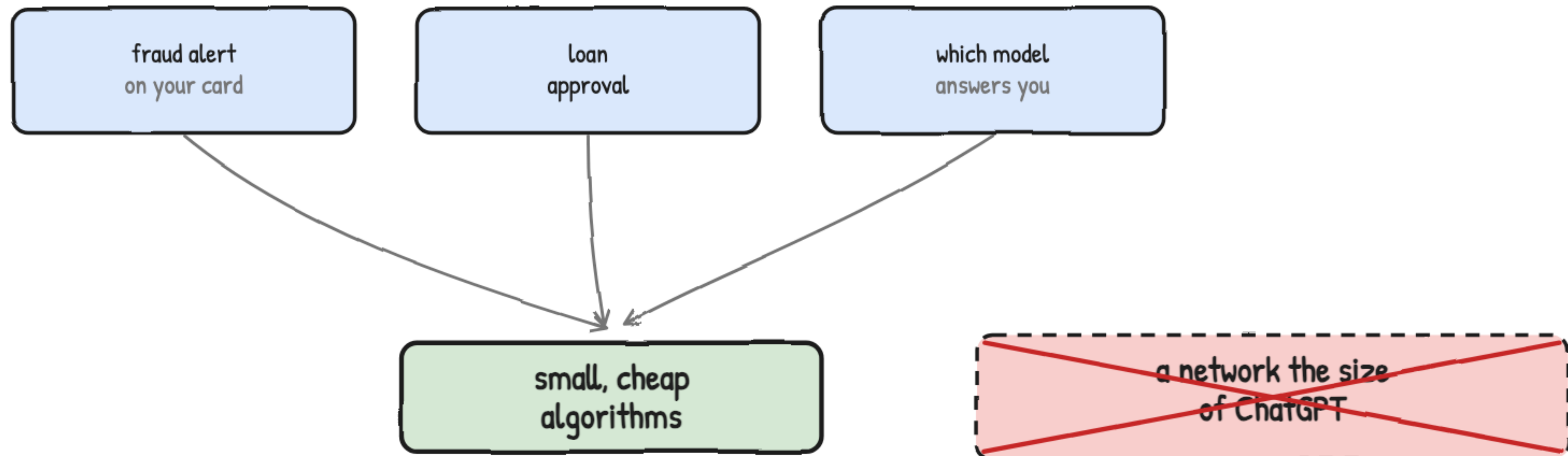
Small, cheap algorithms decide more than you think

Gourav Shah · School of DevOps & AI · Lesson + Lab + Quiz

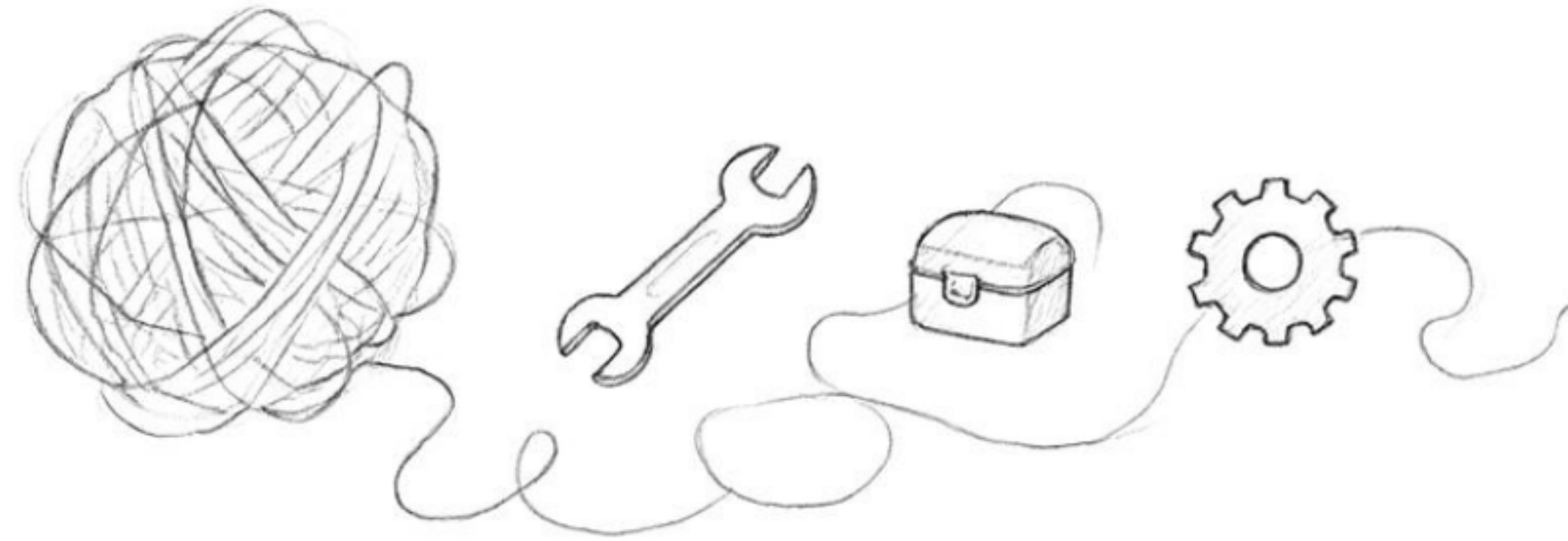


What decided that fraud alert, that loan approval, and which model answered your last prompt?

None of that ran on a model the size of ChatGPT



Small, cheap algorithms decide more than you think



The toolbox, end to end

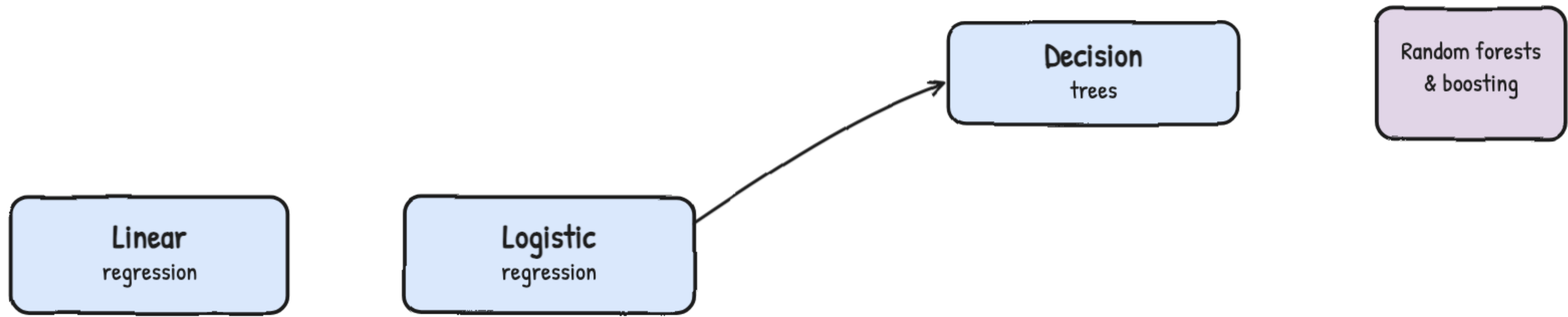
One thread connects every tool in this toolbox

Linear
regression

Logistic
regression

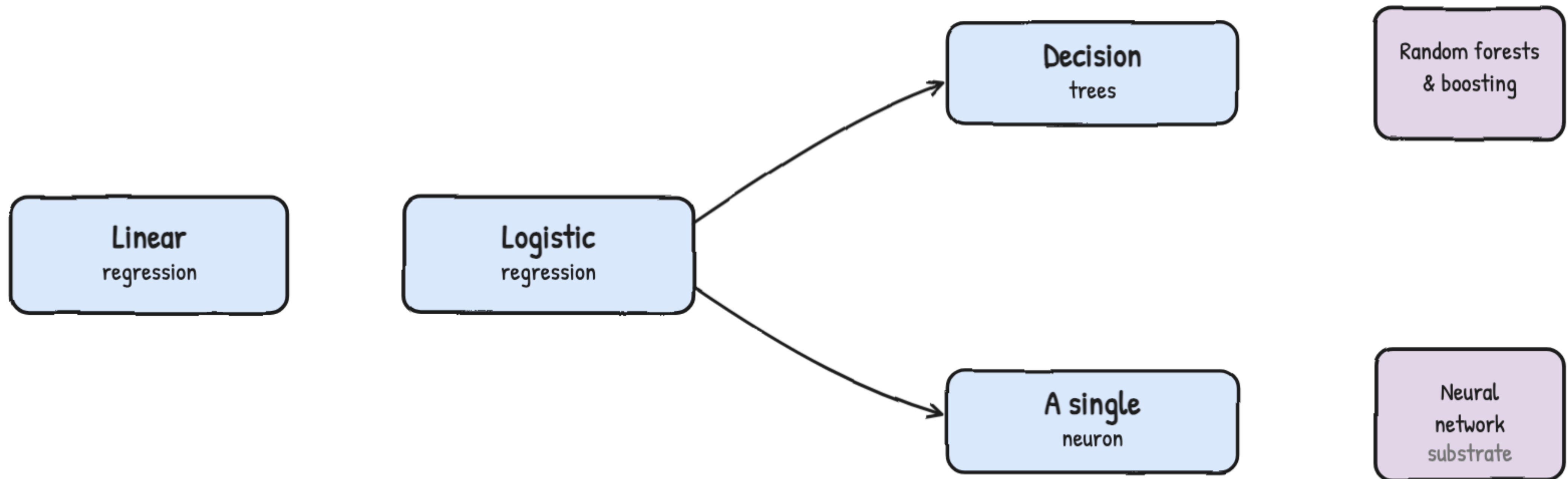
A weighted sum, reused and extended, five times over

One thread connects every tool in this toolbox

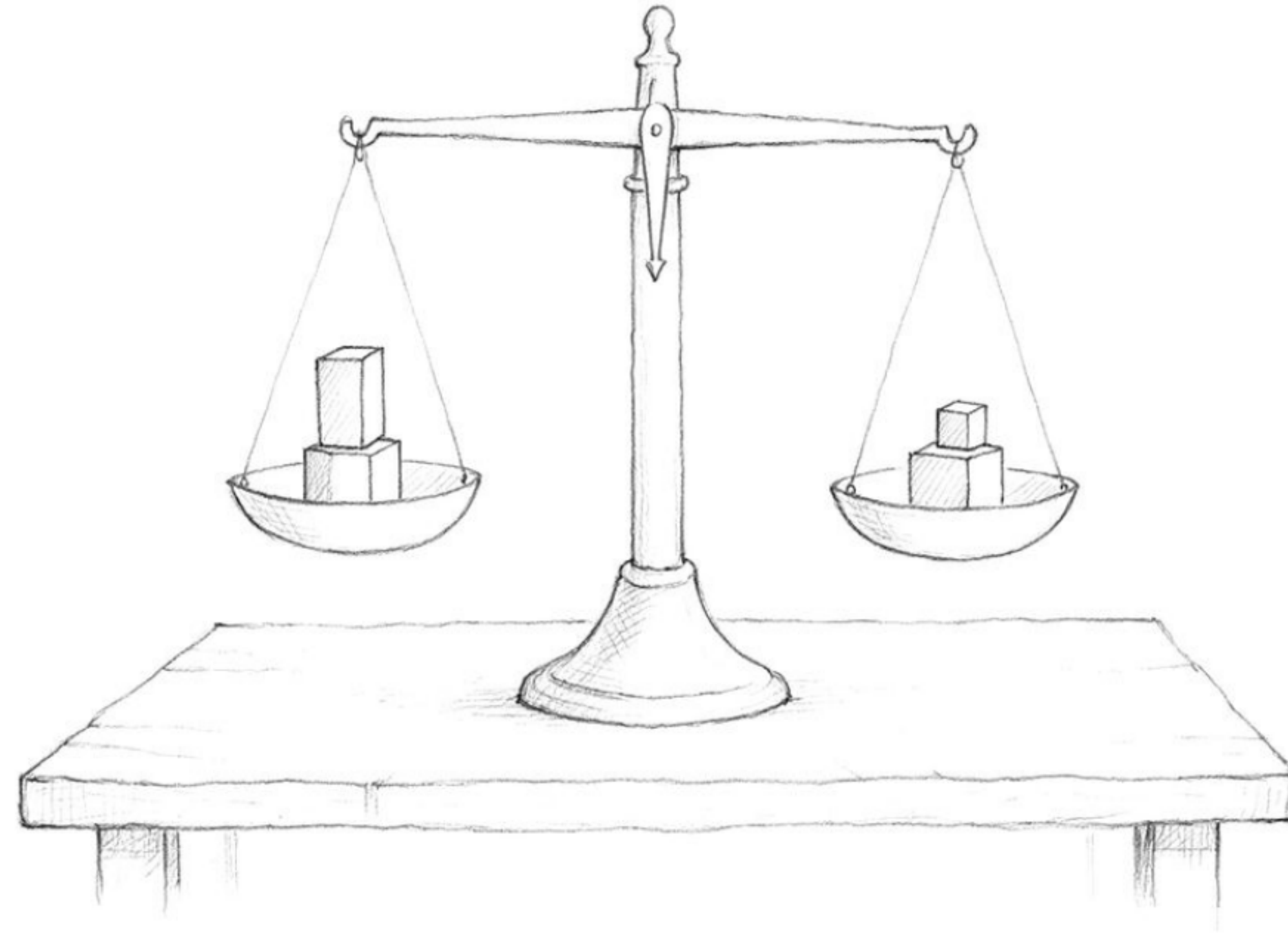


A weighted sum, reused and extended, five times over

One thread connects every tool in this toolbox

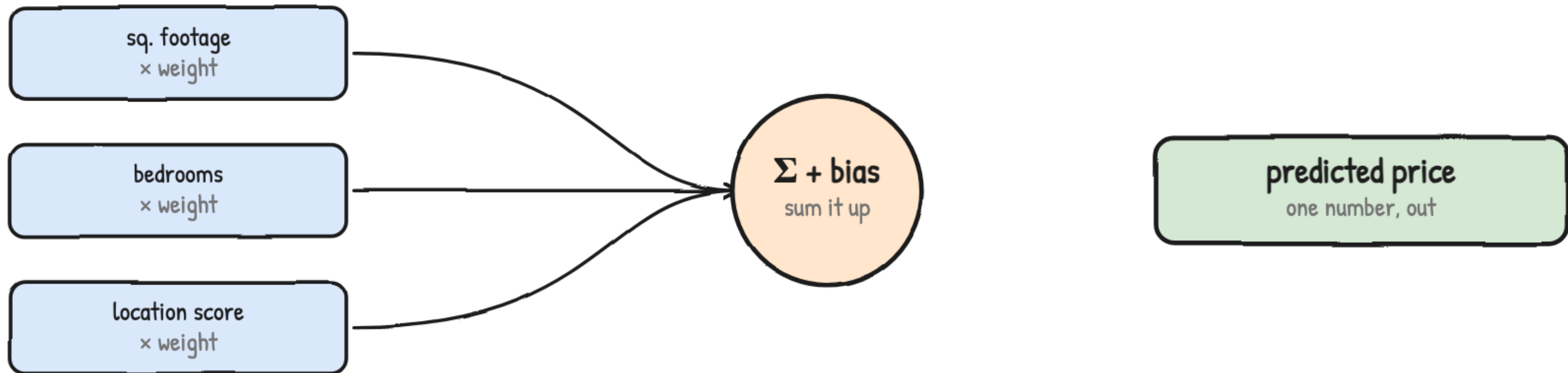


A weighted sum, reused and extended, five times over



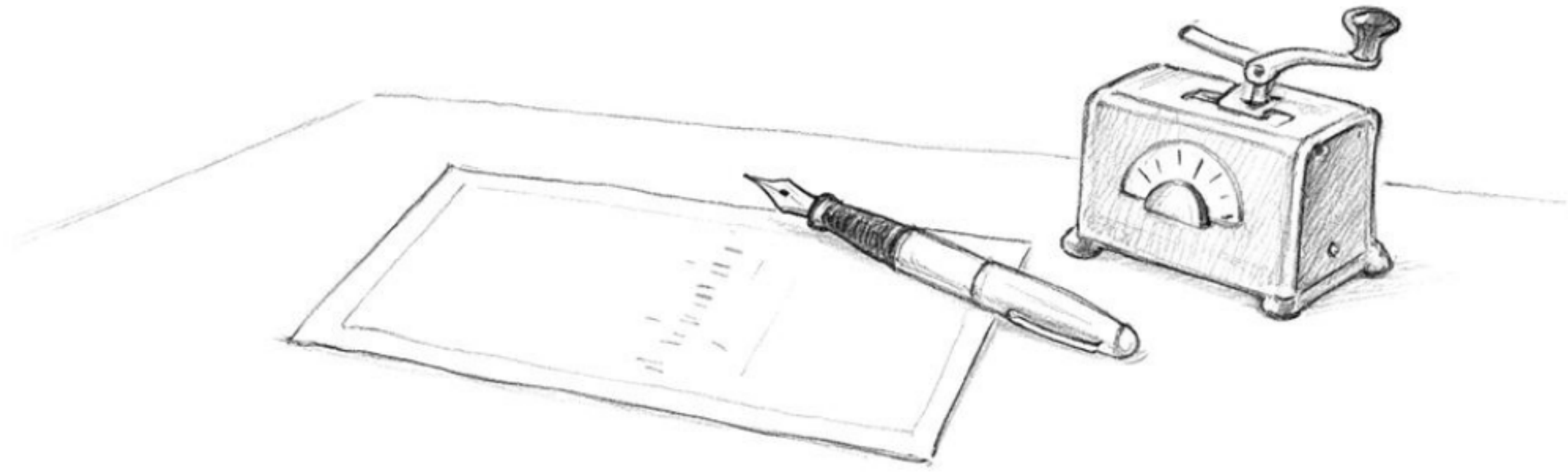
Linear regression: the model you already built

Weight each feature, add them up — you built this in Module 3



Module 3's lab — same arithmetic, real houses

Same arithmetic, real houses



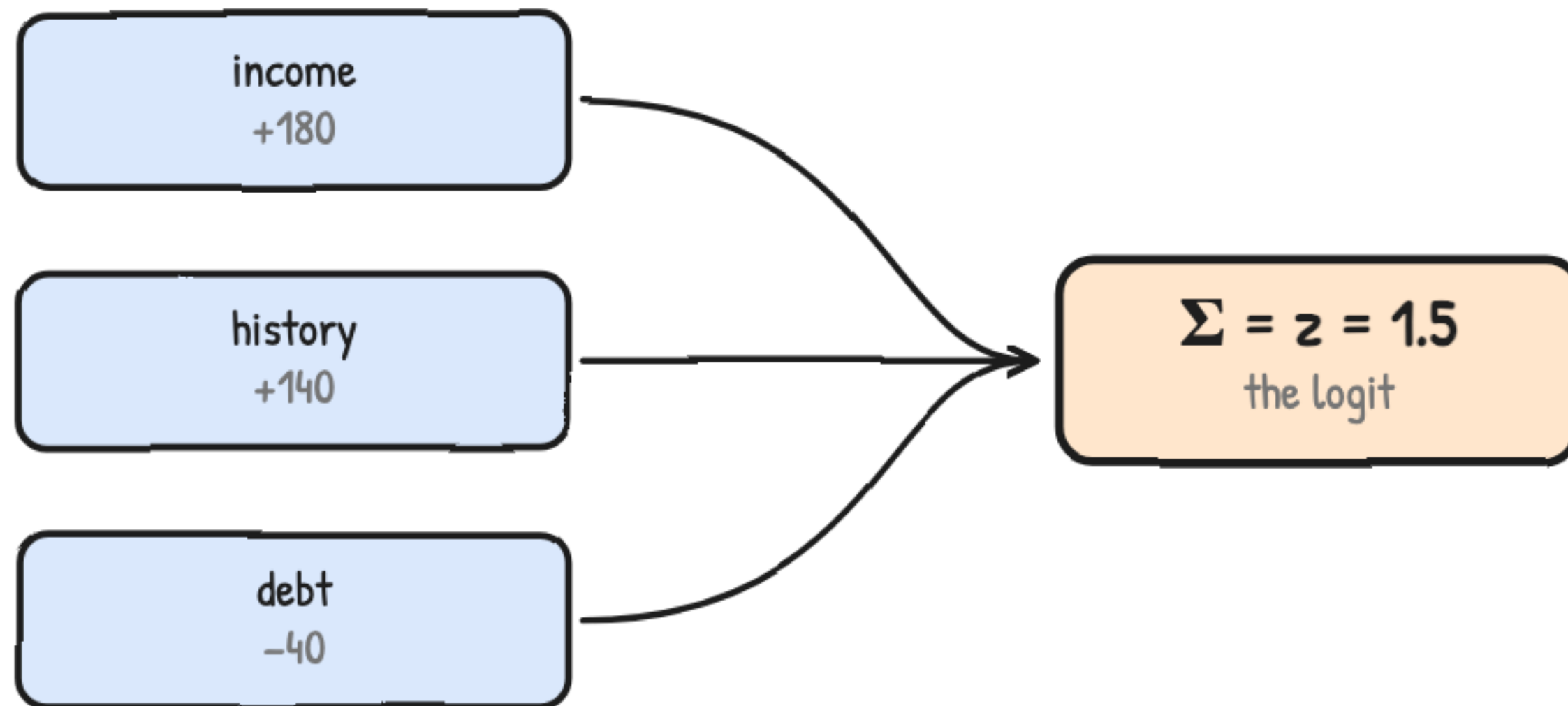
Logistic regression: the logit an LLM computes for every word

A loan officer's scorecard is a weighted sum with a job to do



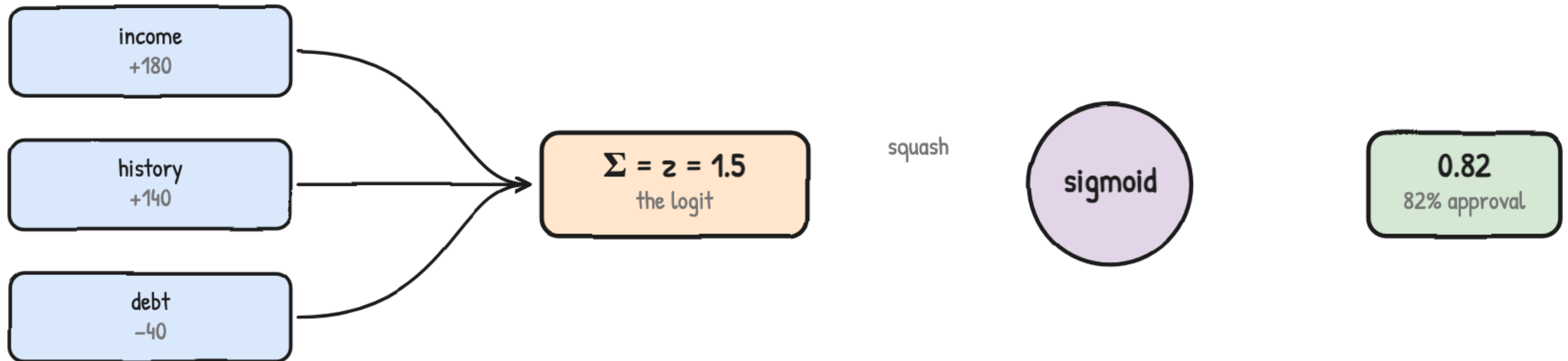
Income adds points, debt subtracts points

Squash the score through sigmoid and the raw sum gets a name: the logit



Remember that word — it is this section's payoff

Squash the score through sigmoid and the raw sum gets a name: the logit



Remember that word — it is this section's payoff

An LLM runs this exact math for every word
it could say next

Same shape, aimed at a vocabulary instead of two outcomes

An LLM runs this exact math for every word it could say next

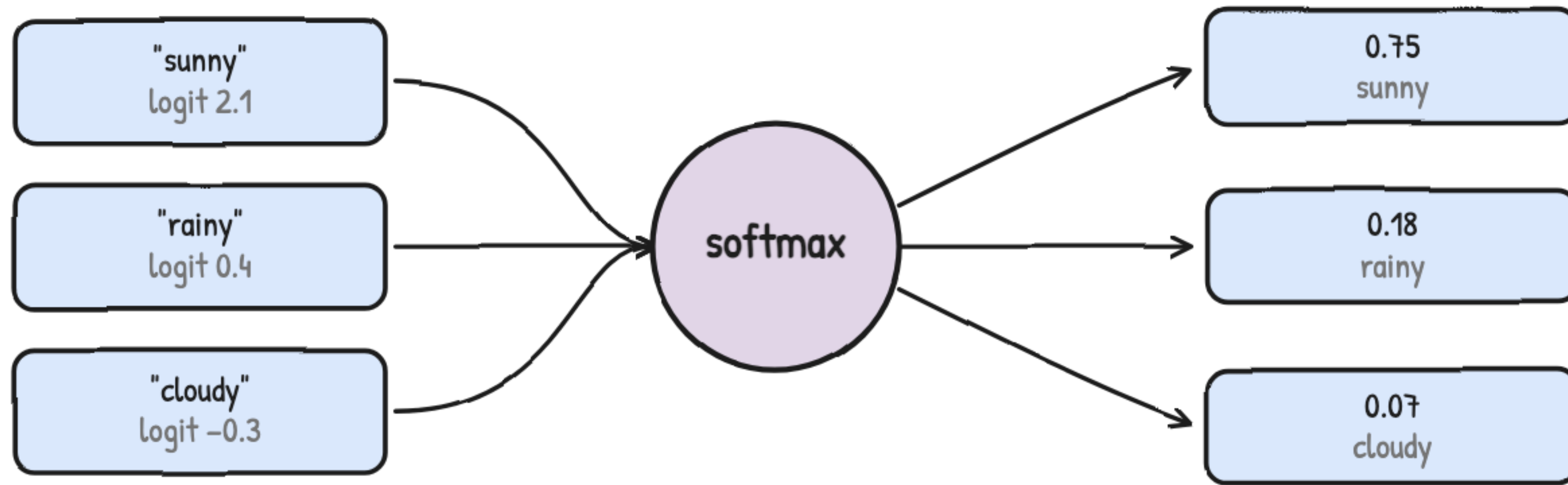
"sunny"
logit 2.1

"rainy"
logit 0.4

"cloudy"
logit -0.3

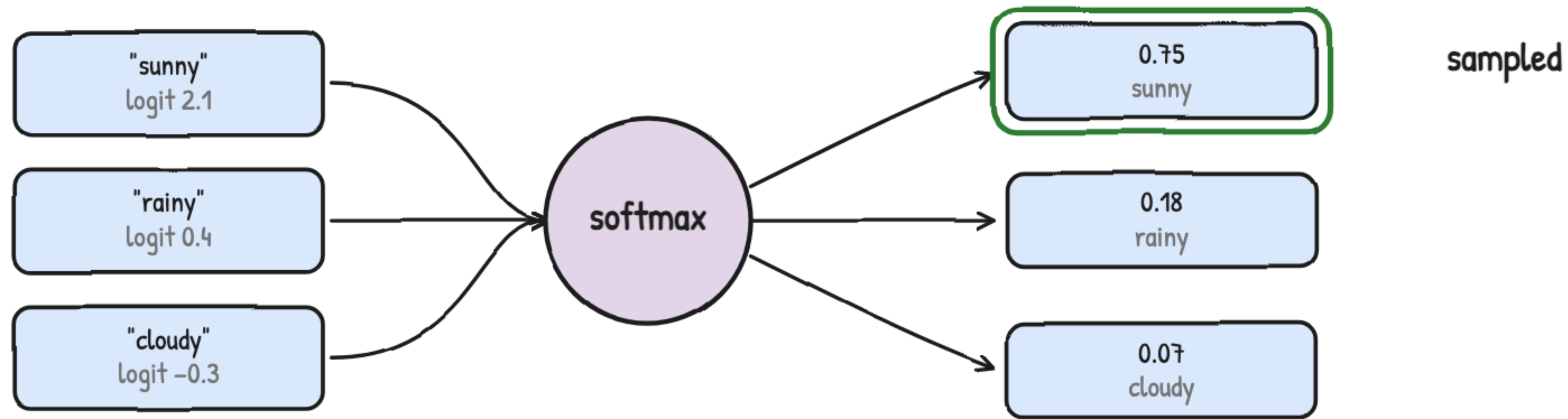
Same shape, aimed at a vocabulary instead of two outcomes

An LLM runs this exact math for every word it could say next



Same shape, aimed at a vocabulary instead of two outcomes

An LLM runs this exact math for every word it could say next

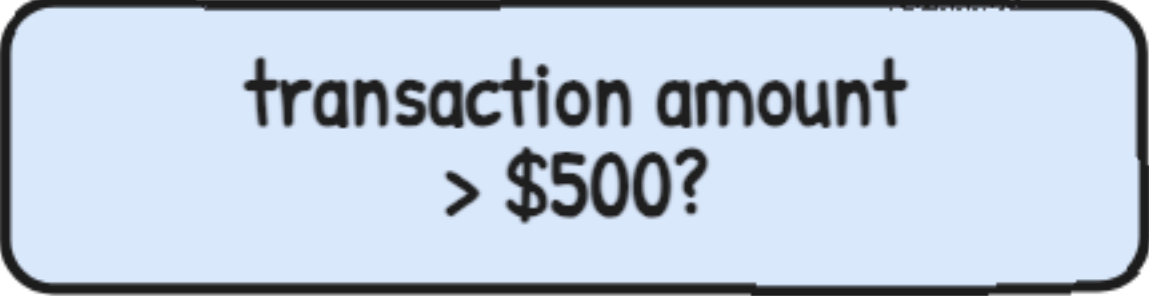


Same shape, aimed at a vocabulary instead of two outcomes



Decision trees: a flowchart that learns itself

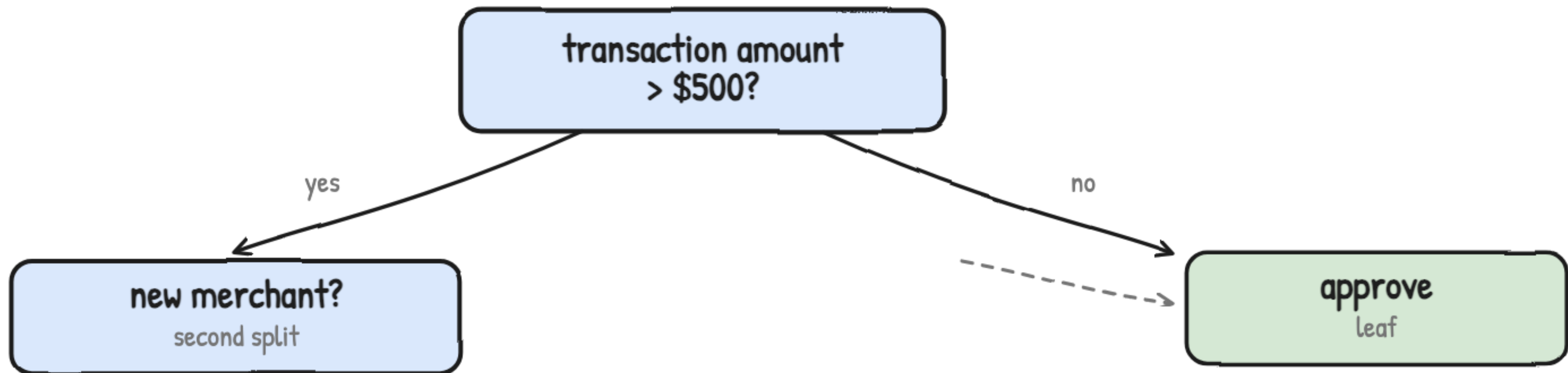
Like a helpdesk flowchart, except the questions are learned from data



transaction amount
> \$500?

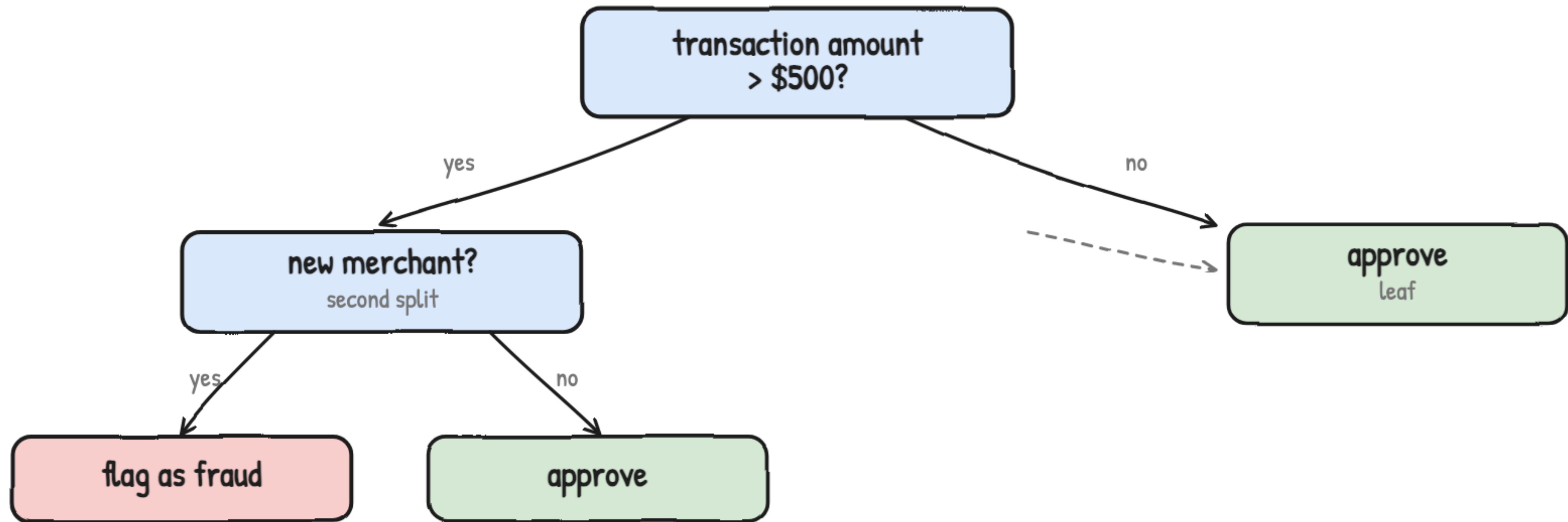
Trace it, question by question, to see why it decided

Like a helpdesk flowchart, except the questions are learned from data



Trace it, question by question, to see why it decided

Like a helpdesk flowchart, except the questions are learned from data

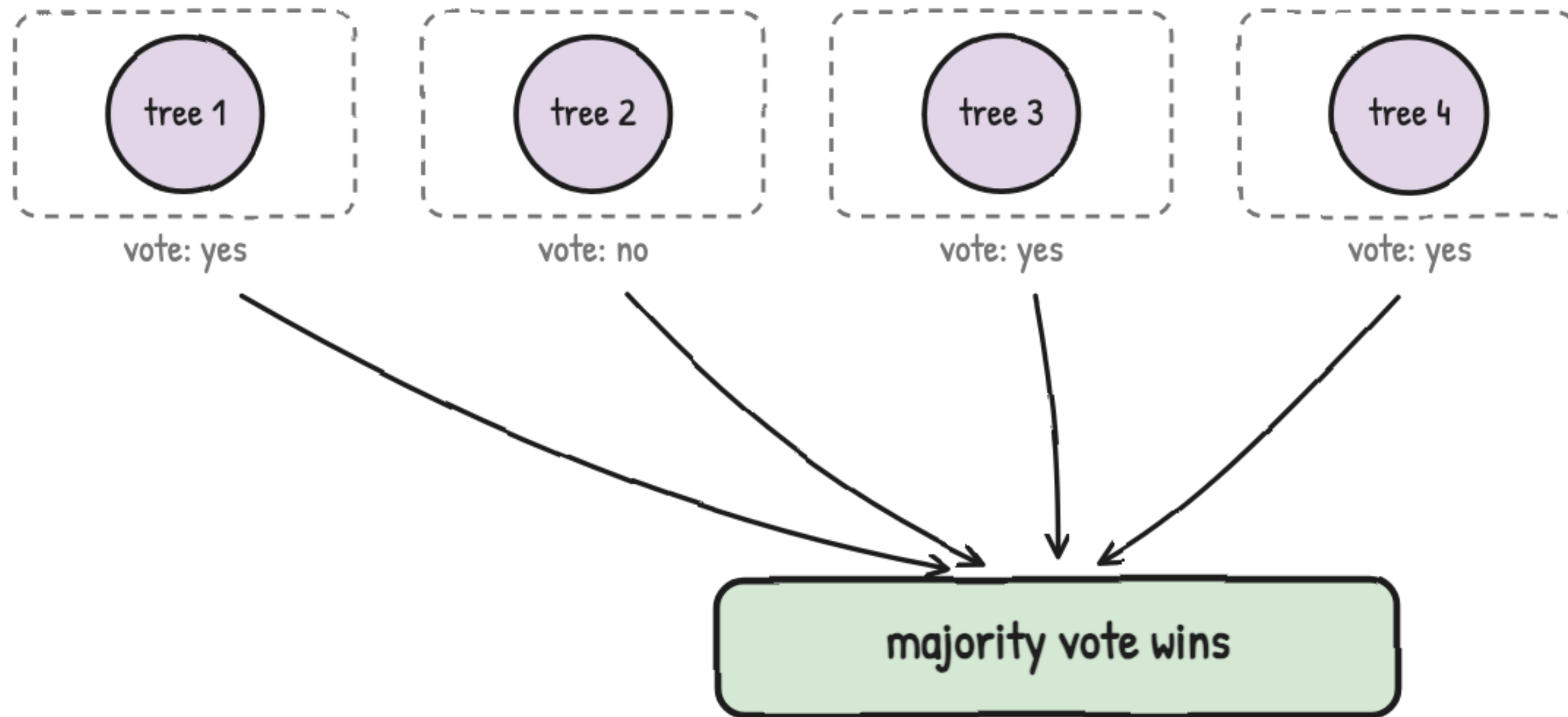


Trace it, question by question, to see why it decided



Random forests and boosting: why tabular data still belongs to trees

A random forest is a panel of judges, not one judge



Many trees, each on a random slice of data, averaged

Boosting is a chain of editors, each one fixing the last pass



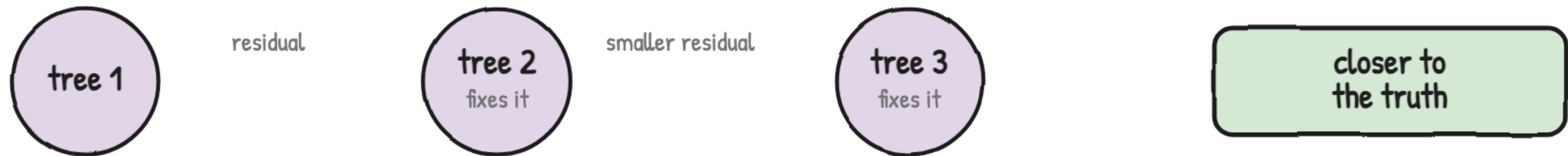
Each new tree is trained to fix what the last one got wrong

Boosting is a chain of editors, each one fixing the last pass



Each new tree is trained to fix what the last one got wrong

Boosting is a chain of editors, each one fixing the last pass



Each new tree is trained to fix what the last one got wrong

On rows and columns, trees still beat the network most of the time

rows & columns

fraud detection

search & ad ranking

model routers

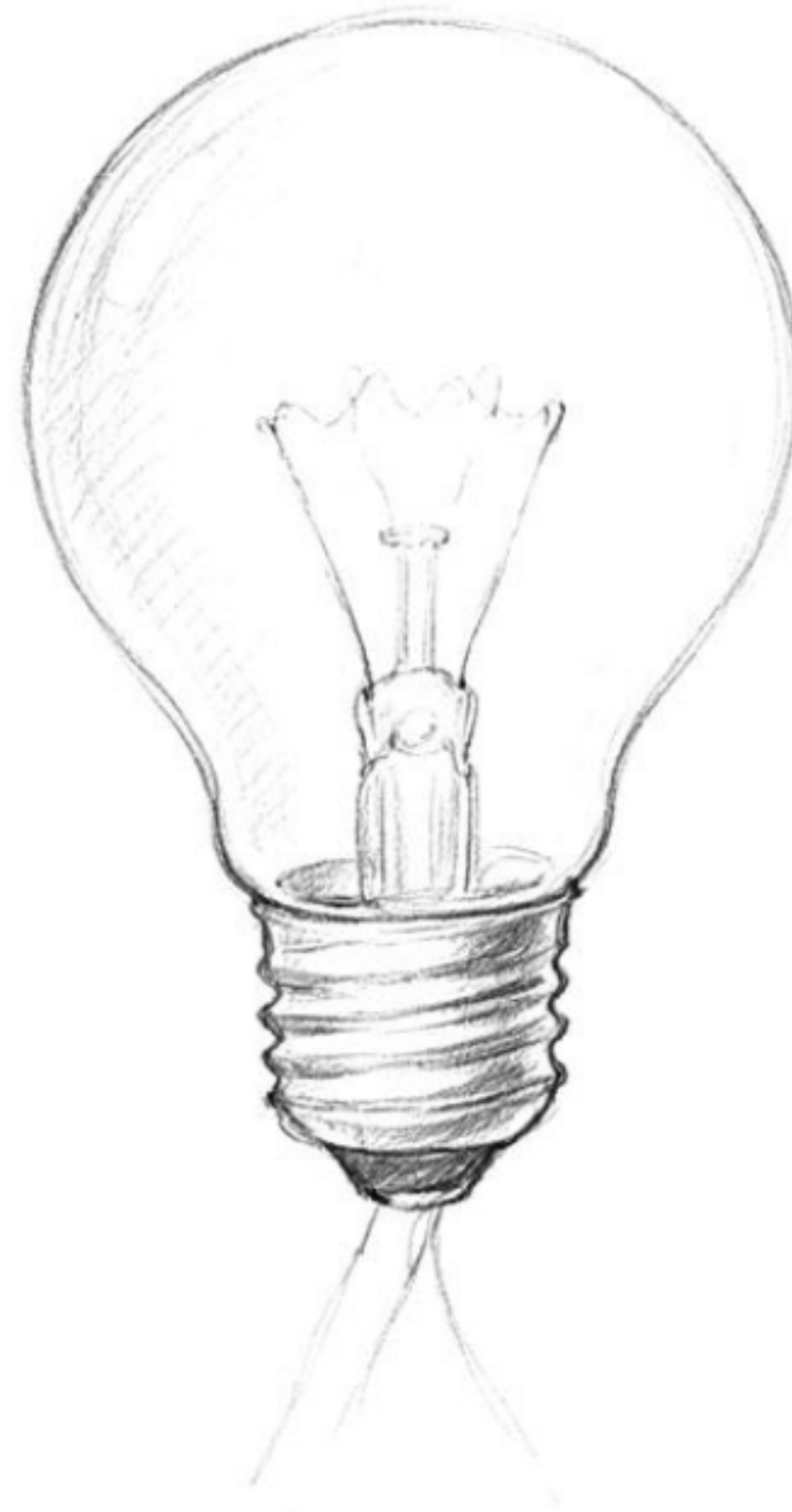
forest / boosting usually wins,
needs far less data

images, audio, raw text

unstructured signal

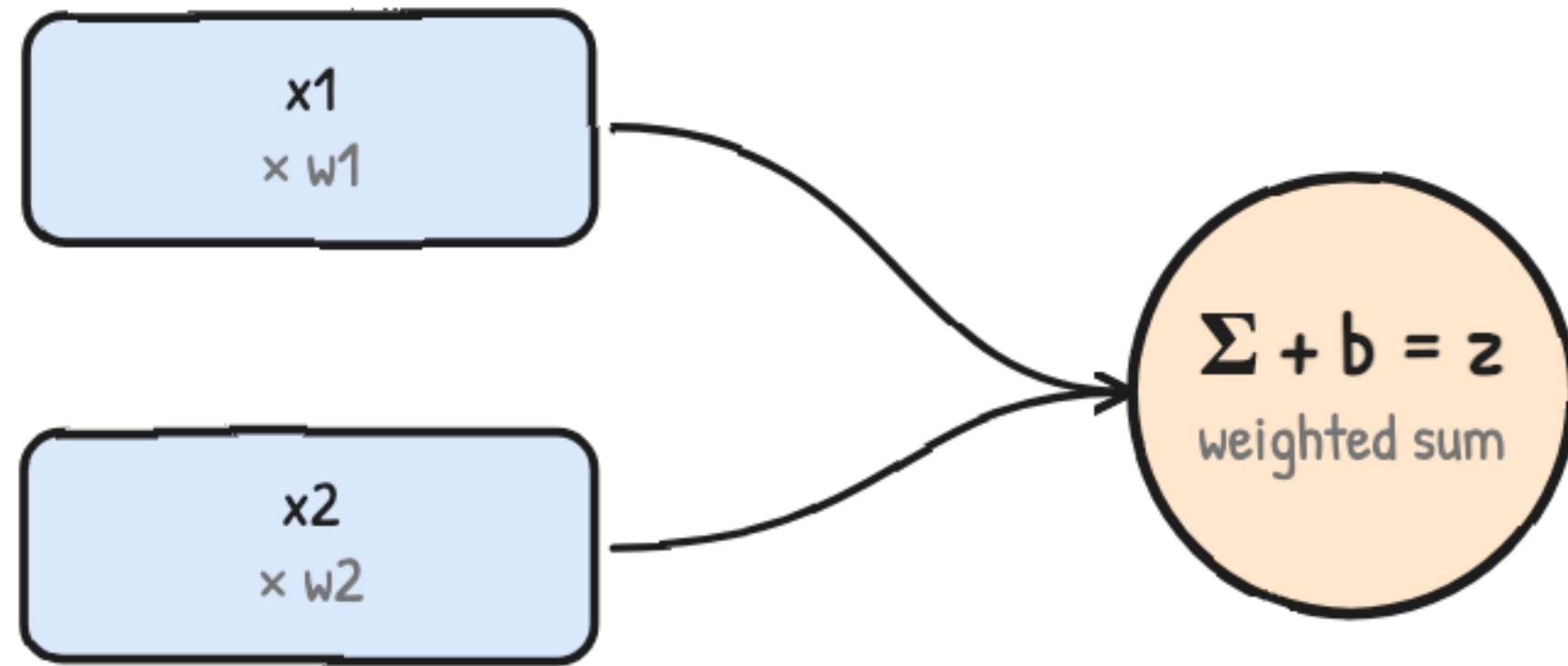
neural network —
learns its own features

Structured data is still tree territory



The neuron: one weighted decision, stacked into a network

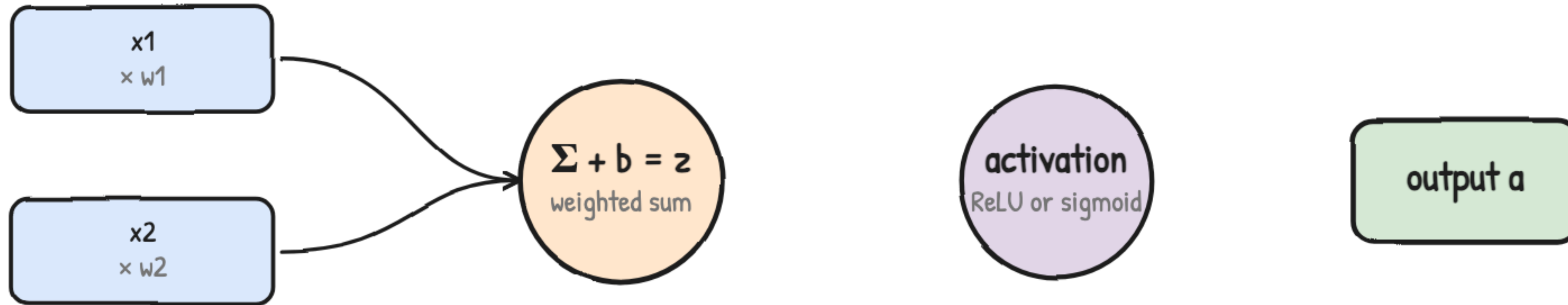
A neuron is logistic regression wearing a different name



same shape as the loan officer's scorecard

Same shape as the last section's scorecard

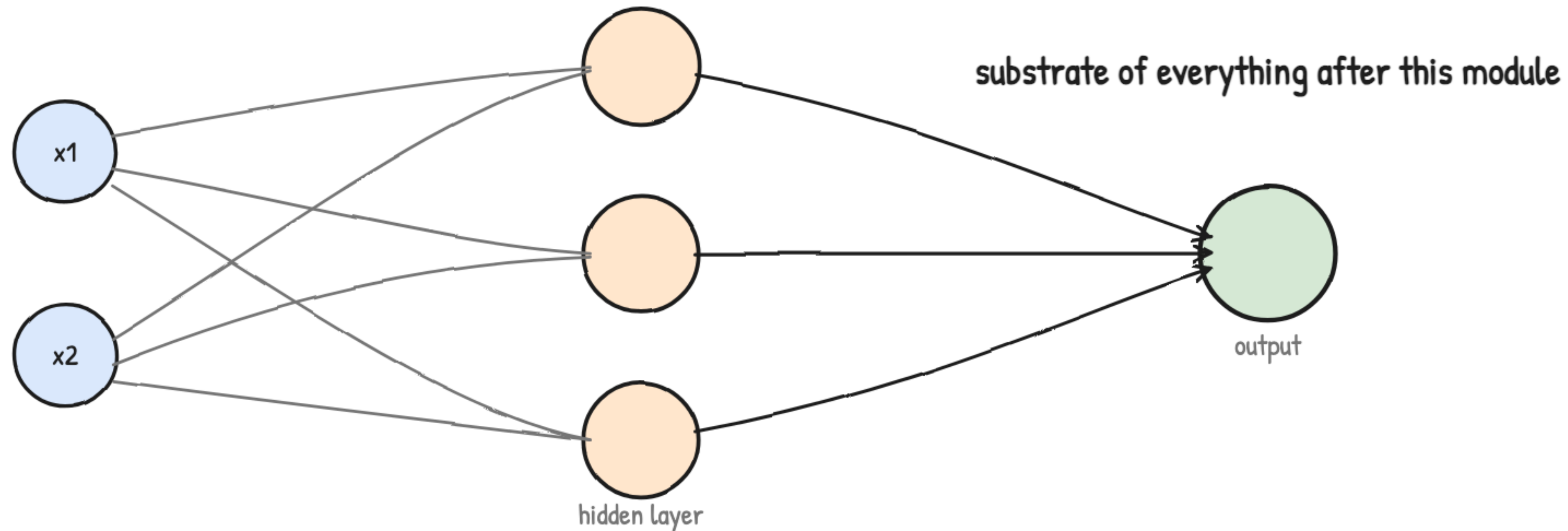
A neuron is logistic regression wearing a different name



same shape as the loan officer's scorecard

Same shape as the last section's scorecard

Stack neurons into layers and it keeps getting more capable



Each layer combines what the last one noticed



Which algorithm do I use?

Match your data's shape to the tool that fits it

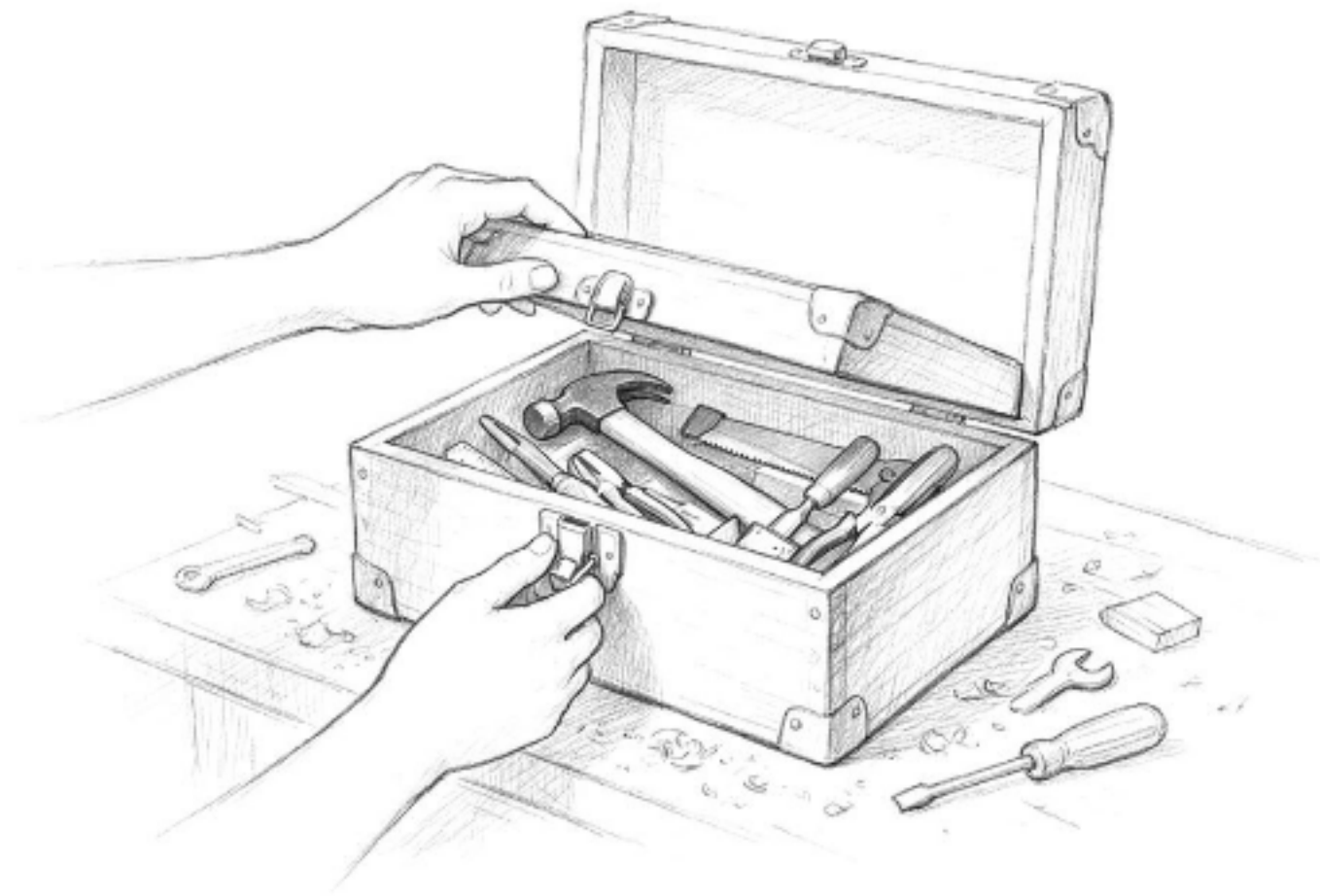
- 1 rows/columns, predict a number
→ linear regression
- 2 rows/columns, predict yes/no
→ logistic regression
- 3 need to explain the reasoning
→ a decision tree
- 4 need the best accuracy you can get
→ random forest or boosting
- 5 images, audio, raw text
→ a neural network

usually wins on tabular data

When unsure, this is the row to reread

MODULE 4 TAKEAWAYS

You now know the arithmetic sitting under an LLM



- 1 linear regression — a weighted sum
- 2 logistic regression — the logit
- 3 decision trees — a learned flowchart
- 4 random forests & boosting — trees, combined
- 5 the neuron — same math, stacked into a network

Now open Module 4 · Lab — Digit Classifier. · Gourav Shah · School of DevOps & AI